



ESP32-S3 DevKitC-1 Pinout Diagram & Safe GPIOs



PANAKOS / 1 DECEMBER 2025 / ESP32 BOARDS & MODULES, GETTING STARTED, SCHEMATICS / 6 COMMENTS

The [ESP32-S3-DevKitC-1](#) has a lot of GPIOs, but not every pin is equally convenient to use.

Some pins affect boot, some are shared with native USB or debugging, and GPIO35–37 may be unavailable depending on the flash and PSRAM configuration of the module fitted to your board.

This guide focuses on [ESP32-S3-DevKitC-1 pinout and GPIO selection](#), so you can connect sensors, relays, displays, SPI devices, UART peripherals and USB hardware without unexpected boot or peripheral conflicts.

The [ESP32-S3-DevKitC-1](#) is Espressif's official entry-level development board for [ESP32-S3](#)-WROOM-1, [ESP32-S3](#)-WROOM-1U and [ESP32-S3](#)-WROOM-2 modules.

The [ESP32-S3](#) combines a dual-core Xtensa LX7 processor with 2.4 GHz Wi-Fi, Bluetooth Low Energy, native USB and a flexible GPIO matrix.

Scope: This guide covers the official [ESP32-S3-DevKitC-1](#) with the J1 and J3 22-pin headers. The board exists in several flash/PSRAM configurations and in more than one hardware revision, which matters for a few GPIOs.

You can compare your board with the [official ESP32-S3-DevKitC-1 documentation](#).



ESP32-S3-DevKitC-1 PINOUT GUIDE

GPIOs, Safe Pins and Wiring Guide for ESP32-S3-DevKitC-1

BOARD REVISIONS

Original DevKitC-1 → RGB LED on **GPIO48**
DevKitC-1 v1.1 → RGB LED on **GPIO38**

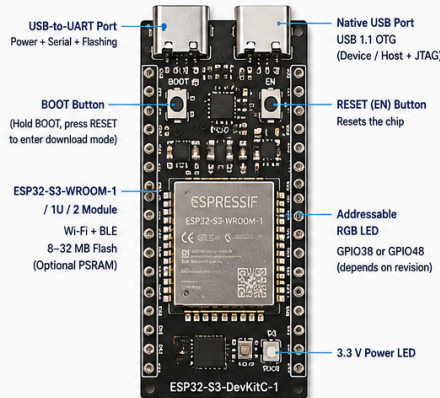
QUICK REFERENCE

- BEST GENERAL-PURPOSE GPIOs**
GPIO1, 2, 4–18, 21, 47
Safe for most digital I/O, PWM, etc.
- USE WITH CAUTION**
GPIO19, 20 (USB), 38/48 (LED),
39–42 (JTAG), 43–44 (UART0)
- AVOID FOR NORMAL I/O**
GPIO0, 3, 45, 46 (strapping pins)
Boot mode / VDD_SPI / JTAG
- MODULE-DEPENDENT PINS**
GPIO35, 36, 37
May be used internally for flash/PSRAM
- TOTAL GPIOs**
45 (GPIO0–21, GPIO26–48)
22–25 do not exist

J1 (Top to Bottom)

ADC1_CH0 / TOUCH1	1	GPIO1
ADC1_CH1 / TOUCH2	2	GPIO2
ADC1_CH2 / TOUCH3	3	GPIO3*
ADC1_CH3 / TOUCH4	4	GPIO4
ADC1_CH4 / TOUCH5	5	GPIO5
ADC1_CH5 / TOUCH6	6	GPIO6
ADC1_CH6 / TOUCH7	7	GPIO7
ADC1_CH7 / TOUCH8	8	GPIO8
ADC1_CH8 / TOUCH9	9	GPIO9
ADC1_CH9 / TOUCH10	10	GPIO10
GPIO	11	GPIO11
GPIO	12	GPIO12
GPIO	13	GPIO13
GPIO	14	GPIO14
GPIO	15	GPIO15
GPIO	16	GPIO16
GPIO	17	GPIO17
GPIO	18	GPIO18
GPIO	29	GPIO19**
GPIO	20	GPIO20**
	22	GND
	22	3V3

ESP32-S3-DEVKITC-1 OVERVIEW



J3 (Top to Bottom)

1	GPIO47	GPIO
2	GPIO48	GPIO / LED (Rev. Original)
3	GPIO21	GPIO
4	GPIO0*	Strapping (BOOT)
5	GPIO45*	Strapping (VDD_SPI)
6	GPIO46*	Strapping (BOOT/ROM)
7	GPIO3*	Strapping (JTAG)
8	GPIO41	MTDI (JTAG)
9	GPIO42	MTMS (JTAG)
10	GPIO39	MTCK (JTAG)
11	GPIO40	MTDO (JTAG)
12	GPIO44	U0RXD (UART0 RX)
13	GPIO43	U0TXD (UART0 TX)
14	GND	Ground
15	5V	5V Input / Output
16	GND	Ground
17	3V3	3.3V Output
18	EN	Chip Enable (High = On)
19	VP	ADC1_CH0 (Input Only)
20	VN	ADC1_CH1 (Input Only)
21	IO46	Reserved (Internal Use)
22	NC	Not Connected

* Strapping Pins – Affect boot mode
** USB Pins – Use for native USB (D- / D+)

POWER OPTIONS (CHOOSE ONE)

- USB-to-UART Port**
Default / Recommended
Power + Serial + Flashing
- Native USB Port**
USB 1.1 OTG
(Device / Host + JTAG)
- 5V + GND**
Input 5V
Onboard LDO to 3.3V
- 3V3 + GND**
Input Regulated 3.3V
Ensure enough current
for Wi-Fi peaks

ADC PINS (12-BIT)

ADC1 (GPIO1–10) Recommended for Wi-Fi		ADC2 (GPIO11–20) Usable, but shared with Wi-Fi	
GPIO1	ADC1_CH0	GPIO11	ADC2_CH0
GPIO2	ADC1_CH1	GPIO12	ADC2_CH1
GPIO4	ADC1_CH3	GPIO13	ADC2_CH2
GPIO5	ADC1_CH4	GPIO14	ADC2_CH3
GPIO6	ADC1_CH5	GPIO15	ADC2_CH4
GPIO7	ADC1_CH6	GPIO16	ADC2_CH5
GPIO8	ADC1_CH7	GPIO17	ADC2_CH6
GPIO9	ADC1_CH8	GPIO18	ADC2_CH7
GPIO10	ADC1_CH9	GPIO19	ADC2_CH8 (USB D-)
		GPIO20	ADC2_CH9 (USB D+)

Input Range: Up to ~2.5–3.0V
(Do not exceed 3.3V)

ADC2 Continuous mode not recommended on S3

I²C, SPI, UART RECOMMENDATIONS

I²C (Default)	SPI (Example - SPI2)	UART (Example)
SDA → GPIO8	CS → GPIO10	TX0 → GPIO43
SCL → GPIO9	MOSI → GPIO11	RX0 → GPIO44
	SCK → GPIO12	(USB-to-UART)
	MISO → GPIO13	UART1 (Example)
		TX1 → GPIO17
		RX1 → GPIO18

Many other pins possible via IO Matrix

Good general-purpose SPI mapping on DevKitC-1

Other UARTs can be mapped to many GPIOs

IMPORTANT NOTES

- Avoid GPIO0, 3, 45, 46 for normal I/O (strapping pins).
- Reserve GPIO19 (D-) and GPIO20 (D+) if using native USB.
- GPIO35–37 may be used internally for Octal flash/PSRAM. Check your module.
- RGB LED is on GPIO48 (original) or GPIO38 (v1.1).
- GPIO43/44 are for the USB-to-UART console.
- Logic level is 3.3V. Do not apply 5V to any GPIO.

Recommended / Safe	Use with Caution
Avoid / Strapping Pins	Module Dependent / Special
Power / Ground	

TIP: When in doubt, start with GPIO1, 2, 4–18, 21 and 47. They are the least likely to cause boot or peripheral conflicts.

More guides, projects and tutorials at ESP32.CO.UK

ESP32-S3 DevKitC-1 GPIO Quick Reference

If you simply want to know which pins to use, start here.

GPIO group	Recommendation
GPIO1, 2, 4–18, 21, 47	Best general-purpose starting points
GPIO0, 3, 45, 46	Avoid for normal peripherals: strapping pins
GPIO19, 20	Reserve when using native USB
GPIO35, 36, 37	Module-dependent: unavailable on several Octal-memory versions
GPIO38	RGB LED on DevKitC-1 v1.1
GPIO48	RGB LED on the original DevKitC-1
GPIO39–42	Usable, but also associated with external JTAG
GPIO43, 44	UART0 / USB-to-UART console

GPIO group	Recommendation
GPIO26–32	Internal flash/PSRAM bus; not broken out for normal use

For a project that needs ordinary digital inputs and outputs, **GPIO1**, **GPIO2**, **GPIO4–18**, **GPIO21** and **GPIO47** are the lowest-drama group to start with.

J1

No.	Name	Type 1	Function
1	3V3	P	3.3 V power supply
2	3V3	P	3.3 V power supply
3	RST	I	EN
4	4	I/O/T	RTC_GPIO4, GPIO4, TOUCH4, ADC1_CH3
5	5	I/O/T	RTC_GPIO5, GPIO5, TOUCH5, ADC1_CH4
6	6	I/O/T	RTC_GPIO6, GPIO6, TOUCH6, ADC1_CH5
7	7	I/O/T	RTC_GPIO7, GPIO7, TOUCH7, ADC1_CH6
8	15	I/O/T	RTC_GPIO15, GPIO15, U0RTS, ADC2_CH4, XTAL_32K_P
9	16	I/O/T	RTC_GPIO16, GPIO16, U0CTS,

No.	Name	Type 1	Function
			ADC2_CH5, XTAL_32K_N
10	17	I/O/T	RTC_GPIO17, GPIO17, U1TXD, ADC2_CH6
11	18	I/O/T	RTC_GPIO18, GPIO18, U1RXD, ADC2_CH7, CLK_OUT3
12	8	I/O/T	RTC_GPIO8, GPIO8, TOUCH8, ADC1_CH7, SUBSPICS1
13	3	I/O/T	RTC_GPIO3, GPIO3, TOUCH3, ADC1_CH2
14	46	I/O/T	GPIO46
15	9	I/O/T	RTC_GPIO9, GPIO9, TOUCH9, ADC1_CH8, FSPIHD, SUBSPIHD
16	10	I/O/T	RTC_GPIO10, GPIO10, TOUCH10, ADC1_CH9, FSPICS0, FSPIIO4, SUBSPICS0
17	11	I/O/T	RTC_GPIO11, GPIO11, TOUCH11, ADC2_CH0, FSPID, FSPIIO5, SUBSPID
18	12	I/O/T	RTC_GPIO12, GPIO12, TOUCH12,

No.	Name	Type 1	Function
			ADC2_CH1, FSPICLK, FSPPIO6, SUBSPICLK
19	13	I/O/T	RTC_GPIO13, GPIO13, TOUCH13, ADC2_CH2, FSPIQ, FSPPIO7, SUBSPIQ
20	14	I/O/T	RTC_GPIO14, GPIO14, TOUCH14, ADC2_CH3, FSPIWP, FSPIDQS, SUBSPIWP
21	5V	P	5 V power supply
22	G	G	Ground

J3

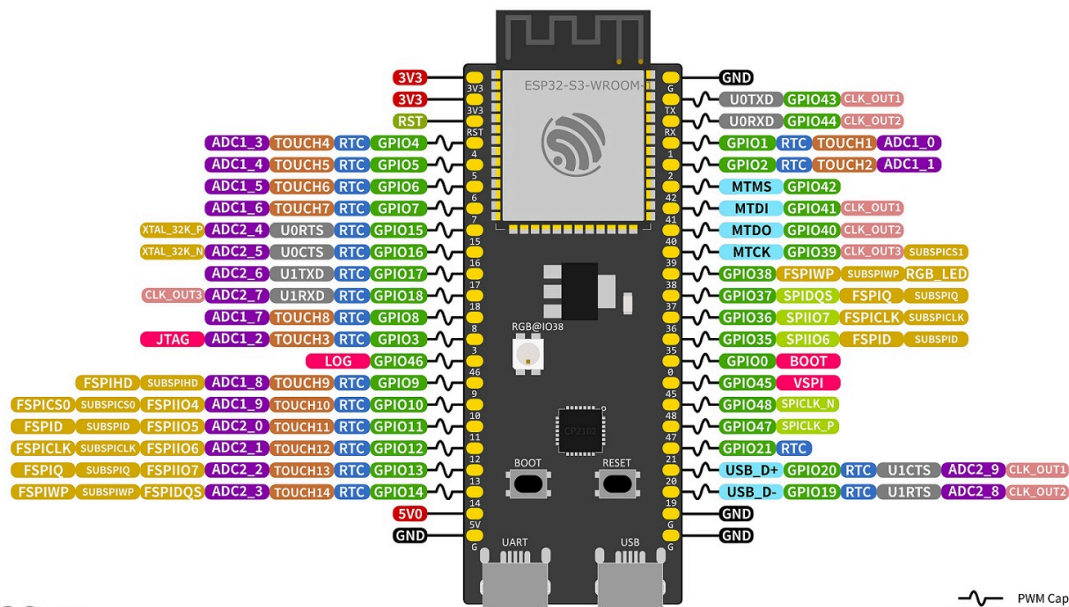
No.	Name	Type	Function
1	G	G	Ground
2	TX	I/O/T	U0TXD, GPIO43, CLK_OUT1
3	RX	I/O/T	U0RXD, GPIO44, CLK_OUT2
4	1	I/O/T	RTC_GPIO1, GPIO1, TOUCH1, ADC1_CH0
5	2	I/O/T	RTC_GPIO2, GPIO2, TOUCH2, ADC1_CH1
6	42	I/O/T	MTMS, GPIO42

No.	Name	Type	Function
7	41	I/O/T	MTDI, GPIO41, CLK_OUT1
8	40	I/O/T	MTDO, GPIO40, CLK_OUT2
9	39	I/O/T	MTCK, GPIO39, CLK_OUT3, SUBSPICS1
10	38	I/O/T	GPIO38, FSPIWP, SUBSPIWP, RGB LED
11	37	I/O/T	SPIDQS, GPIO37, FSPIQ, SUBSPIQ
12	36	I/O/T	SPIIO7, GPIO36, FSPICLK, SUBSPICLK
13	35	I/O/T	SPIIO6, GPIO35, FSPID, SUBSPID
14	0	I/O/T	RTC_GPIO0, GPIO0
15	45	I/O/T	GPIO45
16	48	I/O/T	GPIO48, SPICLK_N, SUBSPICLK_N_DIF F
17	47	I/O/T	GPIO47, SPICLK_P, SUBSPICLK_P_DIF F
18	21	I/O/T	RTC_GPIO21, GPIO21

No.	Name	Type	Function
19	20	I/O/T	RTC_GPIO20, GPIO20, U1CTS, ADC2_CH9, CLK_OUT1, USB_D+
20	19	I/O/T	RTC_GPIO19, GPIO19, U1RTS, ADC2_CH8, CLK_OUT2, USB_D-
21	G	G	Ground
22	G	G	Ground

P: Power supply; I: Input; O: Output; T: High impedance.

ESP32-S3-DevKitC-1



ESP32-S3 Specs

32-bit Xtensa® dual-core @240MHz
 Wi-Fi IEEE 802.11 b/g/n 2.4GHz + BLE 5 Mesh
 512 KB SRAM (16 KB SRAM in RTC)
 384 KB ROM
 45 GPIOs, 4x SPI, 3x UART, 2x I2C,
 14x Touch, 2x I2S, RMT, LED PWM, USB-OTG,
 TWAI®, 2x 12-bit ADC, 1x LCD interface, DVP

MISC Miscellaneous/SPI functions
 CLK_OUTx Clock Output

PWM Capable Pin
 GPIOx GPIO Input and Output
 JTAG/USB JTAG for Debugging and USB
 ADCX_CH Analog-to-Digital Converter
 TOUCHX Touch Sensor Input Channel
 OTHER Other Related Functions
 SERIAL Serial for Debug/Programming
 STRAP Strapping Pin Functions
 RTC RTC Power Domain (VDD3P3_RTC)
 GND Ground
 PWD Power Rails (3V3 and 5V)

Board Overview

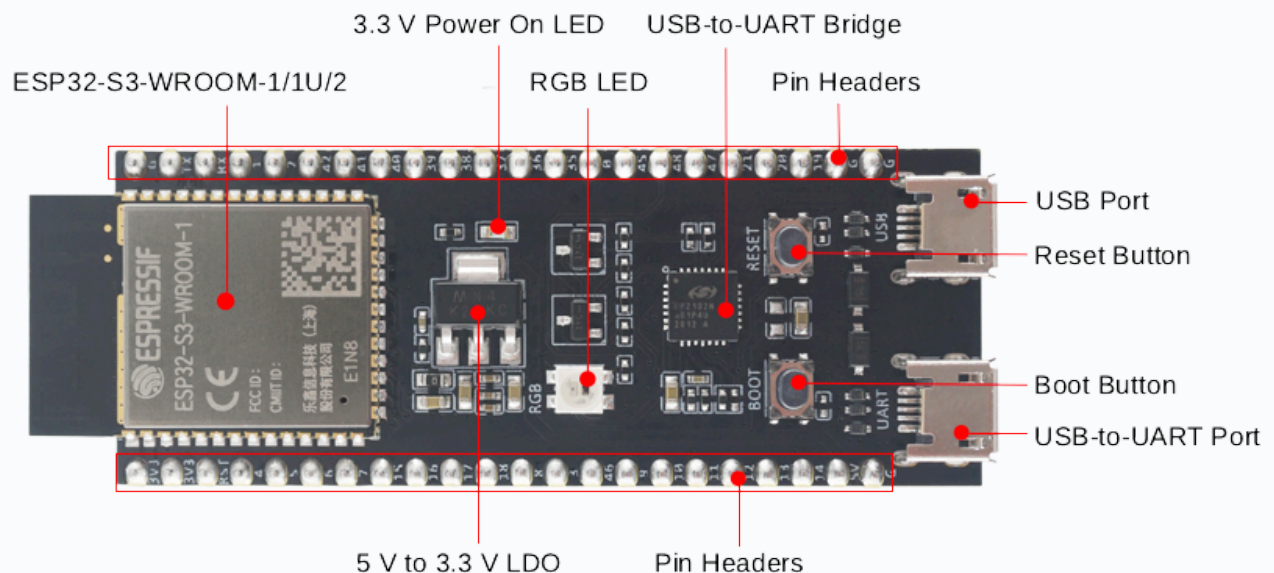
The [ESP32-S3-DevKitC-1](#) exposes most usable [ESP32-S3](#) signals through two 22-pin headers labelled **J1** and **J3**.

The board includes:

- [ESP32](#)-S3-WROOM-1, WROOM-1U or WROOM-2 module
- 5 V to 3.3 V regulator
- USB-to-UART programming port
- native [ESP32](#)-S3 USB port
- BOOT button
- RESET / EN button
- addressable RGB LED
- 3.3 V power LED
- two 22-pin headers

The USB-to-UART port provides the traditional serial programming interface.

The second USB connector is connected directly to the [ESP32](#)-S3's native USB peripheral and can be used for USB devices, USB host applications, firmware upload and built-in USB JTAG.



Watch the RGB LED pin

There are two DevKitC-1 hardware revisions still in circulation.

Original [ESP32-S3-DevKitC-1](#):

- RGB LED → **GPIO48**

[ESP32-S3-DevKitC-1](#) v1.1:

- RGB LED → **GPIO38**

So don't blindly assume GPIO48 controls the LED just because an older tutorial says so.

If you don't know which revision you have, treat **both GPIO38 and GPIO48 as potentially board-connected** until you identify it.

Power Supply Options

There are three basic ways to power the DevKitC-1:

- through the USB ports
- through **5V + GND**
- through **3V3 + GND**

The USB-to-UART and native USB ports form the normal USB power option; Espressif allows either USB connector, or both, to be connected.

For normal development, the easiest option is simply to use the **USB-to-UART connector**.

If you're powering the board from another system:

5V pin

Feed the board with a stable 5 V supply and let its onboard regulator generate 3.3 V.

This is a good option when powering an [ESP32](#) project from something such as a 12 V supply through a DC/DC converter.

3V3 pin

You can bypass the onboard regulator and feed regulated 3.3 V directly.

Your external supply needs enough transient capability for Wi-Fi activity and any peripherals connected to the board.

ESP32-S3 GPIOs are 3.3 V logic

Do **not** connect 5 V signals directly to an [ESP32](#)-S3 GPIO.

Use a level shifter, resistor divider or suitable interface circuit when connecting 5 V hardware.

ESP32-S3 GPIO Basics

The ESP32-S3 has **45 physical GPIOs**:

GPIO0–21 and **GPIO26–48**

GPIO22–25 do not exist.

The important word here is *physical*. Not every chip GPIO is actually available on the DevKitC-1 headers.

The DevKitC-1 breaks out the GPIOs that remain available after the module's flash/PSRAM connections are taken into account.

ESP32-S3 GPIOs are highly multiplexed. Through the GPIO matrix, peripherals such as I²C, SPI, UART, PWM and I²S can be routed to many different pins instead of being permanently tied to one fixed pinout.

That flexibility is one of the S3's biggest advantages, but it also means a pin marked with one peripheral function can often be used for something completely different.

Strapping Pins: GPIO0, GPIO3, GPIO45 and GPIO46

The ESP32-S3 has four **strapping pins**:

- **GPIO0**
- **GPIO3**
- **GPIO45**
- **GPIO46**

Their logic levels are sampled during reset and can affect how the chip starts.

Their functions include:

GPIO	Boot-time function
GPIO0	Boot mode
GPIO46	Boot mode and ROM output behaviour
GPIO45	VDD_SPI configuration
GPIO3	JTAG source selection

GPIO0 is the one most users encounter because pulling it LOW while resetting the ESP32-S3 enters download mode.

That's exactly what the BOOT button does.

After boot, these pins can operate as GPIOs, but attaching external circuitry that holds one of them at the wrong level during reset can produce confusing startup problems.

Recommendation

Unless you're short of pins, **leave GPIO0, GPIO3, GPIO45 and GPIO46 out of ordinary sensor, relay and display wiring.**

There are enough easier GPIOs on the S3 that using strapping pins usually isn't worth the trouble.

Safe GPIOs for Everyday Projects

For a project that must work across the common DevKitC-1 variants, a useful default GPIO pool is:

GPIO1, GPIO2, GPIO4–18, GPIO21 and GPIO47

These avoid:

- the four strapping pins

- native USB
- the module-dependent GPIO35–37 group
- the onboard RGB LED on either board revision
- UART0
- the traditional external JTAG pins

That doesn't mean all the other GPIOs are unusable.

It simply means these pins generally require the least thought.

GPIOs that are usable with conditions

GPIO19 and GPIO20

Perfectly usable as GPIO or ADC inputs **if you are not using native USB**.

If native USB is active, reserve them for USB.

GPIO35, GPIO36 and GPIO37

Whether these pins are available depends on the module fitted to your DevKitC-1.

On boards using ESP32-S3-WROOM modules with certain Octal flash/PSRAM configurations, GPIO35–37 are used internally and cannot be used externally.

If you are writing a guide or designing something intended to work with multiple DevKitC-1 versions, **don't depend on GPIO35–37**.

GPIO38 and GPIO48

One of these drives the addressable RGB LED depending on the board revision.

They can still be useful when you know exactly which DevKitC-1 revision you have.

GPIO39–42

These pins can be GPIOs, but they are also the traditional JTAG signals:

- GPIO39 → MTCK
- GPIO40 → MTDO

- GPIO41 → MTDI
- GPIO42 → MTMS

If you don't use external JTAG debugging, they become useful additional GPIOs.

GPIO43 and GPIO44

These are the default UART0 console signals:

- **GPIO43 → TX0**
- **GPIO44 → RX0**

They are connected to the board's USB-to-UART interface.

You can repurpose them, but doing so can interfere with serial logging, programming workflows or anything connected through the USB-to-UART bridge.

ADC Pins

The ESP32-S3 has two 12-bit SAR ADC units providing 20 analog channels.

ADC1

ADC1 uses **GPIO1–10**:

GPIO	ADC channel
GPIO1	ADC1_CH0
GPIO2	ADC1_CH1
GPIO3	ADC1_CH2
GPIO4	ADC1_CH3
GPIO5	ADC1_CH4
GPIO6	ADC1_CH5
GPIO7	ADC1_CH6
GPIO8	ADC1_CH7

GPIO	ADC channel
GPIO9	ADC1_CH8
GPIO10	ADC1_CH9

GPIO3 is also a strapping pin, so I would normally use **GPIO1**, **GPIO2** or **GPIO4–10** first.

ADC2

ADC2 uses **GPIO11–20**:

GPIO	ADC channel
GPIO11	ADC2_CH0
GPIO12	ADC2_CH1
GPIO13	ADC2_CH2
GPIO14	ADC2_CH3
GPIO15	ADC2_CH4
GPIO16	ADC2_CH5
GPIO17	ADC2_CH6
GPIO18	ADC2_CH7
GPIO19	ADC2_CH8
GPIO20	ADC2_CH9

ADC2 is shared with the Wi-Fi subsystem.

Current ESP-IDF handles ADC2 oneshot access better than older ESP32 software did, but ADC2 remains more constrained when Wi-Fi is active, and ADC2 continuous-mode sampling is not recommended on the ESP32-S3.

Best practical choice for analog sensors

For a normal Wi-Fi project, use **ADC1 on GPIO1–10** where possible.

My first choices would be:

GPIO1, GPIO2, GPIO4, GPIO5, GPIO6 or GPIO7

GPIO8–10 are also excellent, although you may want to reserve GPIO8/9 for I²C and GPIO10 for SPI chip select if you follow the mappings later in this guide.

Never allow the analog input to exceed the ESP32-S3's 3.3 V supply domain.

For accurate voltage measurements, also remember that an ESP32 ADC is not a laboratory ADC: use calibration if accuracy matters.

Capacitive Touch Pins

The ESP32-S3 has **14 capacitive touch channels on GPIO1–14**.

GPIO	Touch channel
GPIO1	TOUCH1
GPIO2	TOUCH2
GPIO3	TOUCH3
GPIO4	TOUCH4
GPIO5	TOUCH5
GPIO6	TOUCH6
GPIO7	TOUCH7
GPIO8	TOUCH8
GPIO9	TOUCH9
GPIO10	TOUCH10
GPIO11	TOUCH11
GPIO12	TOUCH12
GPIO13	TOUCH13

GPIO	Touch channel
GPIO14	TOUCH14

For a straightforward touch input, I would avoid the GPIO3 strapping pin and start with:

GPIO4–14

Touch inputs can also be used as wake-up sources in low-power projects.

Native USB Pins

Unlike the original ESP32, the ESP32-S3 includes native full-speed USB OTG.

The USB data pins are fixed:

- **GPIO19 → USB D-**
- **GPIO20 → USB D+**

These lines are connected to the ESP32-S3 USB connector on the DevKitC-1.

If you're using native USB for something such as:

- USB CDC serial
- HID keyboard or mouse
- MIDI
- USB storage
- USB host
- USB Serial/JTAG

treat GPIO19 and GPIO20 as **reserved USB pins**.

Do not attach ordinary sensors or outputs to them while using native USB.

If native USB isn't being used, GPIO19 and GPIO20 can be repurposed as normal GPIO or ADC2 inputs.

I²C Pins

The ESP32-S3 does not force you to use one fixed pair of pins for I²C.

The GPIO matrix lets you route SDA and SCL to many GPIOs.

For the DevKitC-1, a clean default is:

Function	GPIO
SDA	GPIO8
SCL	GPIO9

These pins avoid boot, USB, UART and flash/PSRAM complications.

Arduino example:

```
#include <Wire.h>

void setup() {
  Wire.begin(8, 9);  // SDA, SCL
}

void loop() {
}
```

That gives you a predictable I²C connection for devices such as a [BME280](#), OLED display, RTC, [INA219](#) or other sensor.

You can choose other GPIOs if GPIO8/9 are already needed elsewhere.

SPI Pins

SPI is another area where the S3's GPIO matrix provides a lot of freedom.

A particularly sensible general-purpose SPI2 mapping on the DevKitC-1 is:

SPI signal	GPIO
CS	GPIO10
MOSI	GPIO11
SCK	GPIO12
MISO	GPIO13

This group is much better as a general recommendation than GPIO35–38 because it works across the common DevKitC-1 memory variants.

It is suitable for:

- TFT displays
- SD card modules
- Ethernet controllers
- ADC/DAC chips
- RF modules
- SPI sensors

Additional control signals such as display DC, RESET and backlight can then use GPIO14–16 or another free GPIO.

For example:

Display function	GPIO
CS	GPIO10
MOSI	GPIO11
SCK	GPIO12
MISO	GPIO13
DC	GPIO14
RESET	GPIO15
Backlight	GPIO16

Because the signals can be routed through the GPIO matrix, these aren't mandatory pins—they are simply a very convenient layout.

UART and Serial Pins

The ESP32-S3 has multiple UART controllers, and their signals can also be remapped.

On the DevKitC-1, UART0 is used for the normal serial console:

- **GPIO43 → UART0 TX**
- **GPIO44 → UART0 RX**

For that reason, I recommend leaving GPIO43 and GPIO44 alone unless you deliberately want to repurpose the console.

For a second serial peripheral, GPIO17 and GPIO18 form a very convenient pair:

Function	GPIO
UART1 TX	GPIO17
UART1 RX	GPIO18

This is useful for:

- GPS modules
- RS-485 transceivers
- energy meters
- other microcontrollers
- serial displays
- GSM/LTE modules

Arduino example:

```
HardwareSerial DeviceSerial(1);

void setup() {
```

```
DeviceSerial.begin(9600, SERIAL_8N1, 18, 17);  
}
```

Remember that Arduino's `begin()` arguments are **RX first, then TX**.

PWM Pins

The ESP32-S3 LED PWM controller can be routed through the GPIO matrix, so most ordinary output-capable GPIOs can be used for PWM.

For LEDs, fan control, dimming and similar applications, pick from the normal safe GPIO pool rather than wasting a special-function pin.

Good examples include:

GPIO4, GPIO5, GPIO6, GPIO7, GPIO14, GPIO15, GPIO16, GPIO17, GPIO18, GPIO21 or GPIO47

For motors, relays and other inductive loads, the GPIO should drive a proper transistor, MOSFET or driver circuit rather than powering the load directly.

Practical ESP32-S3-DevKitC-1 Pin Recipes

Wi-Fi sensor with I²C and a relay

A simple arrangement that leaves all problematic pins alone:

Function	Recommended GPIO	Notes
I ² C SDA	GPIO8	Sensors / OLED
I ² C SCL	GPIO9	Sensors / OLED
Relay	GPIO21	Simple digital output
Button / reed switch	GPIO4	Digital input
Analog sensor	GPIO1 or GPIO2	ADC1

This leaves native USB, UART0, strapping pins and flash/PSRAM-related pins untouched.

SPI TFT display

A convenient mapping is:

Function	GPIO
CS	GPIO10
MOSI	GPIO11
SCK	GPIO12
MISO	GPIO13
DC	GPIO14
RESET	GPIO15
Backlight	GPIO16

That gives you a neat sequential group on the J1 side of the board.

Native USB device with button and status indication

Function	GPIO
USB D-	GPIO19
USB D+	GPIO20
User button	GPIO4
Extra status LED	GPIO5

For the onboard RGB LED, check your board revision first:

- original DevKitC-1 → **GPIO48**
- DevKitC-1 v1.1 → **GPIO38**

UART device plus I²C sensors

Function	GPIO
UART RX	GPIO18
UART TX	GPIO17
I ² C SDA	GPIO8
I ² C SCL	GPIO9
Digital output	GPIO21

This combination keeps UART0 free for debugging and programming.

Pins I Would Avoid First

When starting a new DevKitC-1 project, I would not choose these pins unless there is a specific reason.

GPIO0, GPIO3, GPIO45 and GPIO46

Strapping pins. They can work after startup, but external circuitry can interfere with boot.

GPIO19 and GPIO20

Keep free when using native USB.

GPIO35, GPIO36 and GPIO37

Availability depends on the module's flash/PSRAM configuration.

Don't design around them unless you have checked the exact module fitted to your board.

GPIO38 and GPIO48

One is connected to the RGB LED depending on the DevKitC-1 revision.

GPIO43 and GPIO44

UART0 console and USB-to-UART interface.

GPIO39–42

Fine as GPIO when you're not using traditional external JTAG, but I would use the completely free pins first.

ESP32-S3-DevKitC-1 Rules of Thumb

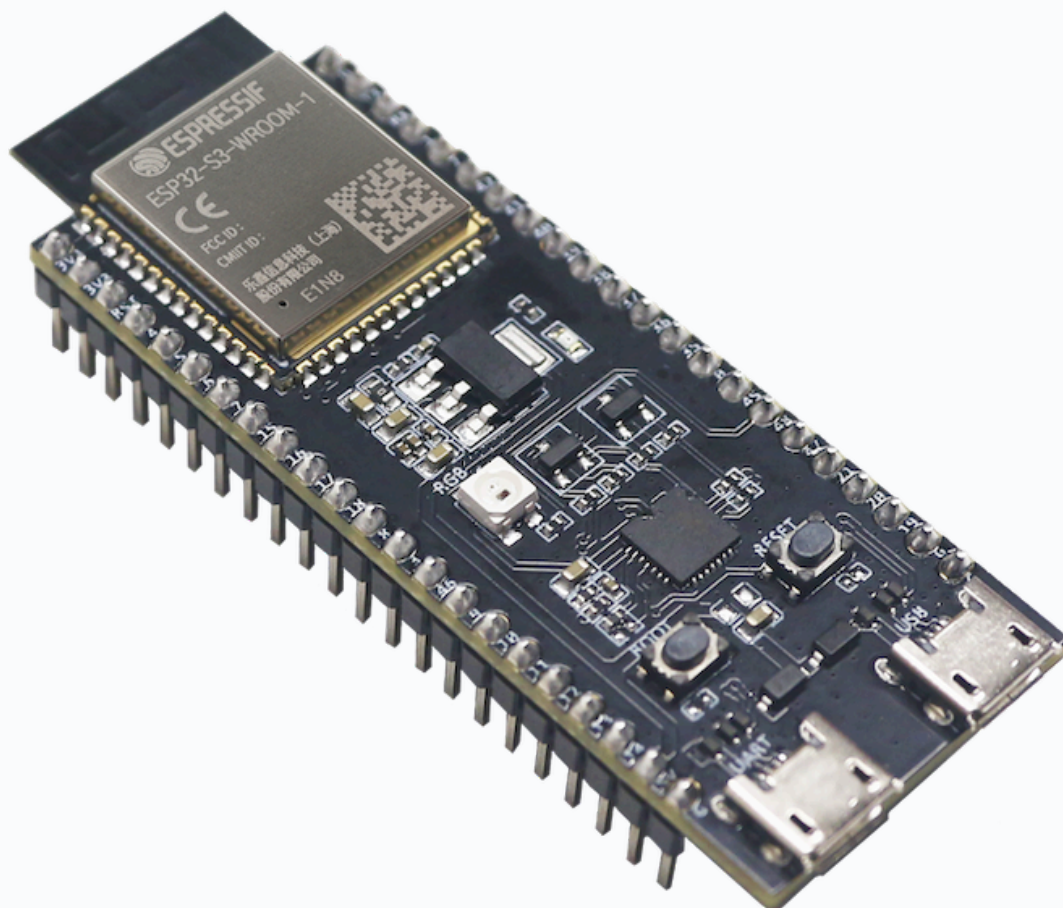
The [ESP32-S3-DevKitC-1](#) is an extremely flexible development board once you separate the genuinely free GPIOs from pins that have additional jobs.

The practical rules are simple:

- **Start with GPIO1, 2, 4–18, 21 and 47 for ordinary I/O**
- **Avoid GPIO0, 3, 45 and 46 for normal peripherals**
- **Reserve GPIO19 and 20 when using native USB**
- **Don't assume GPIO35–37 are available without checking your exact module**
- **Check whether your RGB LED is on GPIO38 or GPIO48**
- **Keep GPIO43/44 free if you want the normal UART console**
- **Prefer ADC1 on GPIO1–10 for Wi-Fi projects**
- **GPIO8/9 make a convenient I²C pair**
- **GPIO10–13 make a convenient SPI bus**
- **GPIO17/18 make a convenient additional UART**

Follow those rules and the [ESP32-S3-DevKitC-1](#) becomes a very predictable platform for **Wi-Fi sensors, Home Assistant nodes, relays, displays, native USB devices and general ESP32-S3 development.**

The important thing isn't memorising every alternate peripheral function printed on a pinout diagram. It is knowing **which GPIOs have hidden responsibilities and choosing the uncomplicated pins first.**



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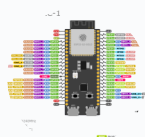
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